EXPERIMENT-6

Building an Anonymous FTP Scanner Using Python

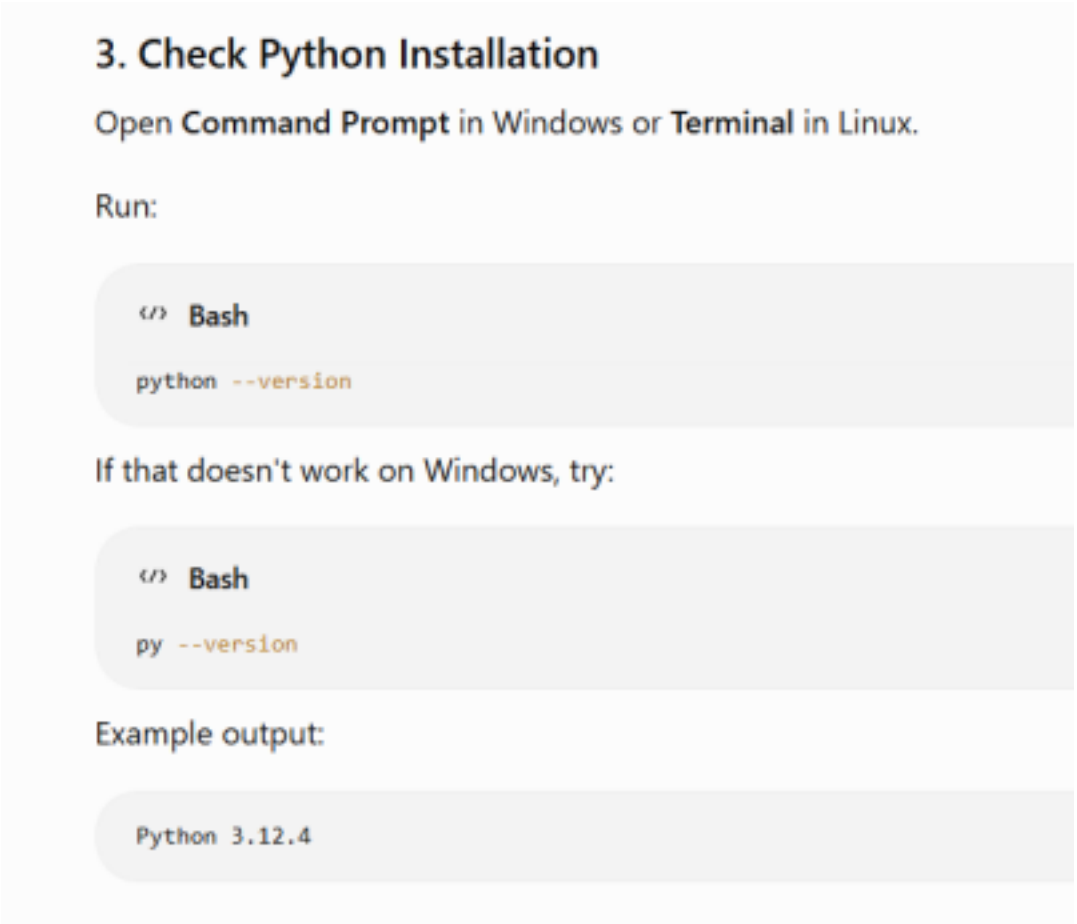
Aim

To develop a Python-based anonymous FTP scanner using the built-in ftplib module and determine whether an FTP server permits anonymous access.

Requirements

- Windows/Linux computer
- Python 3.x
- Python's built-in ftplib module
- An FTP server that you are authorized to test
- Internet/network connection if testing a remote server

No separate installation of ftplib is required because it is included with Python.



4. Create the Python File

Create a file named:

```
anonymous_ftp_scanner.py
```

Open it in IDLE, VS Code, Notepad++, or another Python editor.

5. Import the `ftplib` Module

Start with:

```
Python
from ftplib import FTP
```

`FTP` provides functions for connecting and communicating with an FTP server.

6. Write the Scanner Program

```
from ftplib import FTP

from ftplib import all_errors


def anonymous_ftp_scan(host, port=21):

 print("=" * 50)

 print(" ANONYMOUS FTP SCANNER")

 print("=" * 50)


 print(f"Target: {host}")

 print(f"Port : {port}")

 print()


 ftp = FTP()


 try:
print("[1] Connecting to FTP server...")


 ftp.connect(host, port, timeout=10)
```

```
print("[+] FTP server is reachable")
print("[+] Server response:", ftp.getwelcome())

print("\n[2] Testing anonymous login...")

    ftp.login(user="anonymous", passwd="anonymous@example.com")

print("[+] Anonymous login SUCCESSFUL")
print("[!] The FTP server permits anonymous access.")

print("\n[3] Checking accessible directory...")

files = ftp.nlst()

if files:
print("[+] Directory listing is available:")
for item in files:
print(" ", item)
else:
print("[-] Directory is empty or listing is unavailable.")

ftp.quit()

except all_errors as error:
print("[-] Anonymous login failed or FTP connection failed.")
print("[!] Error:", error)

try:
ftp.close()
except:
```

pass

Enter an authorized FTP server here

target = input("Enter FTP server address: ")

anonymous_ftp_scan(target)

8. Run the Program

Open Command Prompt/Terminal in the folder containing the Python file.

Run:

```bash
python anonymous_ftp_scanner.py
```

You will see:

```
------------------------------------------------
       ANONYMOUS FTP SCANNER
------------------------------------------------
Target:
Port  : 21

Enter FTP server address:
```

Enter an **authorized** FTP server.

For a local FTP lab server, for example:

```
127.0.0.1
```

9. Example Successful Output

```
================================================
       ANONYMOUS FTP SCANNER
================================================
Target: 127.0.0.1
Port  : 21

[1] Connecting to FTP server...
[+] FTP server is reachable
[+] Server response: 220 FTP Server Ready

[2] Testing anonymous login...
[+] Anonymous login SUCCESSFUL
[!] The FTP server permits anonymous access.

[3] Checking accessible directory...
[+] Directory listing is available:
     welcome.txt
     public
     readme.txt
```

10. Example Failed Output

```
[1] Connecting to FTP server...
[+] FTP server is reachable

[2] Testing anonymous login...
[-] Anonymous login failed or FTP connection failed.
[!] Error: 530 Login incorrect.
```